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COMPREHENSIVE FORMULAS & FUNCTIONS HANDBOOK [EXCEL & GOOGLE SHEETS]

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PART 1: FORMULA BASICS, OPERATORS & CELL REFERENCING

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1. FORMULA SYNTAX RULES:

- Every formula and function MUST start with an equals sign (=).
- Functions follow the pattern: =FUNCTION_NAME(argument1, argument2, ...)
- Arguments are separated by commas (,).
- Text strings inside formulas must be wrapped in quotes: "Delayed".
- Numerical values and cell references do NOT use quotes: A1 + 50.

2. ARITHMETIC OPERATORS & PEMDAS PRECEDENCE:

+	Addition	=A1 + B1
-	Subtraction	=A1 - B1
*	Multiplication	=A1 * B1
/	Division	=A1 / B1
^	Exponentiation	=A1 ^ 2 (A1 squared)
%	Percentage	=A1 * 18% (Multiplies by 0.18)
()	Parentheses	Controls calculation order: =(A1 + B1) * C1

* PEMDAS Order: Parentheses -> Exponents -> Multiply/Divide -> Add/Subtract

3. CELL REFERENCING TYPES (CRITICAL LAB & EXAM SKILL):

- Relative Reference (A1):
  Shifts row and column automatically when copied or dragged.
  Example: In C1, =A1*B1. Dragged down to C2 automatically becomes =A2*B2.
- Absolute Reference (A1):
  Locks BOTH column and row with $. Never shifts when copied anywhere.
  Shortcut Key: Press F4 (or Fn + F4) while editing a cell reference.
  Example: =B2 * D1 (where D1 holds a fixed Tax Rate or Aviation Fuel Surcharge).
- Mixed Reference ($A1 or A$1):
  $A1 locks Column A, while row changes when dragged down.
  A$1 locks Row 1, while column changes when dragged across.

4. ERROR CODES & DIAGNOSTIC FIXES:

- #DIV/0! : Attempted division by zero or empty cell. Fix: Wrap in IFERROR.
- #N/A : Value not available (Lookup found no matching record).
- #VALUE! : Wrong data type passed (e.g. attempting arithmetic on text).
- #REF! : Invalid cell reference (a referenced row or column was deleted).
- #NAME? : Misspelled function name (e.g. =SUMM(A1:A10) instead of =SUM).
- #### : Column is too narrow to display number. Fix: Double-click header.

PART 2: ESSENTIAL STATISTICAL FUNCTIONS:

=SUM(range)	: Adds all numbers in range	=SUM(H2:H51)
=AVERAGE(range)	: Calculates arithmetic mean	=AVERAGE(L2:L51)
=MIN(range)	: Returns smallest value in range	=MIN(I2:I51)
=MAX(range)	: Returns largest value in range	=MAX(I2:I51)
=COUNT(range)	: Counts cells with NUMBERS only	=COUNT(L2:L51)
=COUNTA(range)	: Counts all NON-EMPTY cells	=COUNTA(A2:A51)
=COUNTBLANK(range)	: Counts empty cells in range	=COUNTBLANK(M2:M51)
=ROUND(num, digits)	: Rounds to specified decimals	=ROUND(A1, 1)
=MOD(num, divisor)	: Returns division remainder	=MOD(125, 60) -> 5

PART 3: LOGICAL, COMPARISON & CONDITIONAL FUNCTIONS

COMPARISON OPERATORS:
= (Equal), <> (Not Equal), > (Greater Than), < (Less Than), >= (GTE), <= (LTE)

1. IF FUNCTION (BINARY DECISION):
Syntax: =IF(logical_test, value_if_true, value_if_false)
Example: =IF(L2 > 15, "Delayed", "On-Time")
Evaluates flight delay: if L2 > 15 returns "Delayed", else "On-Time".

2. NESTED IF (MULTI-TIER DECISION HIERARCHY):
Syntax: =IF(cond1, val1, IF(cond2, val2, val3))
Example: =IF(L2=0, "On-Time", IF(L2<=30, "Minor Delay", "Major Disruption"))

3. IFS FUNCTION (MODERN CLEAN MULTI-CONDITION SYNTAX):
Syntax: =IFS(condition1, val1, condition2, val2, [TRUE, default_val])
Example: =IFS(L2=0, "On-Time", L2<=15, "Short Buffer", TRUE, "Delayed")

4. LOGICAL OPERATORS: AND, OR, NOT
- =AND(cond1, cond2): Returns TRUE only if ALL conditions are TRUE.
  Example: =IF(AND(F2="International", I2>250), "Widebody Gate", "Narrowbody")
- =OR(cond1, cond2): Returns TRUE if AT LEAST ONE condition is TRUE.
  Example: =IF(OR(K2="Delayed", K2="Cancelled"), "Irregular Ops", "Normal")
- =NOT(cond): Reverses boolean truth value.

5. ERROR SUPPRESSION: IFERROR & IFNA
Syntax: =IFERROR(calculation, fallback_value)
Example: =IFERROR(I2 / H2, 0) -> Prevents #DIV/0! if capacity is empty.

PART 4: CRITERIA-BASED AGGREGATIONS (COUNTIF, SUMIF, AVERAGEIF)

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1. COUNTIF & COUNTIFS:
- Single Condition: =COUNTIF(range, criteria)
  Example: =COUNTIF(B2:B51, "IndiGo") -> Counts IndiGo movements
  Example: =COUNTIF(L2:L51, ">30") -> Flights delayed >30 mins
- Multiple Conditions: =COUNTIFS(criteria_range1, crit1, crit_range2, crit2)
  Example: =COUNTIFS(B2:B51, "Air India", K2:K51, "Delayed")

2. SUMIF & SUMIFS:
- Single Condition: =SUMIF(criteria_range, criteria, [sum_range])
  Example: =SUMIF(C2:C51, "T1", J2:J51) -> Sums baggage pieces for T1
- Multiple Conditions: =SUMIFS(sum_range, crit_range1, crit1, crit_range2, crit2)
  * CRITICAL: In SUMIFS, sum_range comes FIRST, not last!
  Example: =SUMIFS(I2:I51, B2:B51, "IndiGo", C2:C51, "T1")

3. AVERAGEIF & AVERAGEIFS:
- Single Condition: =AVERAGEIF(criteria_range, criteria, [avg_range])
  Example: =AVERAGEIF(B2:B51, "Vistara", L2:L51) -> Avg delay for Vistara
- Multiple Conditions: =AVERAGEIFS(avg_range, crit_range1, crit1, ...)
  Example: =AVERAGEIFS(L2:L51, C2:C51, "T2", F2:F51, "International")

PART 5: TEXT MANIPULATION & DATA CLEANING FUNCTIONS

FUNCTION	SYNTAX & DESCRIPTION	EXAMPLE
& (Ampersand)	Concatenation operator Joins text without a formula.	=A2 & " - " & D2 -> "FL-101 - BLR-DEL"
CONCATENATE	=CONCATENATE(text1, text2, ...)	=CONCATENATE(B2, " ", C2)
TEXTJOIN	=TEXTJOIN(delim, ignore_empty, range) Joins array with a custom delimiter.	=TEXTJOIN(", ", TRUE, B2:B6) -> "IndiGo, Air India"
UPPER	=UPPER(text)	=UPPER("spicejet") -> "SPICEJET"
LOWER	=LOWER(text)	=LOWER("AIR INDIA") -> "air india"
PROPER	=PROPER(text)	=PROPER("akasa air") -> "Akasa Air"
TRIM	=TRIM(text) Removes extra leading & trailing spaces.	=TRIM(" BLR ") -> "BLR"
LEN	=LEN(text) Counts total character count in cell.	=LEN("FL-101") -> 6
LEFT	=LEFT(text, num_chars) Extracts n characters from the left.	=LEFT("BLR-BOM", 3) -> "BLR"
RIGHT	=RIGHT(text, num_chars) Extracts n characters from the right.	=RIGHT("BLR-BOM", 3) -> "BOM"
MID	=MID(text, start_num, num_chars) Extracts characters from the middle.	=MID("FL-101", 4, 3) -> "101"
FIND	=FIND(find_text, within_text) Returns character position (Case-Sensitive).	=FIND("-", "BLR-DEL") -> 4
SEARCH	=SEARCH(find_text, within_text)	Case-insensitive position search.
SUBSTITUTE	=SUBSTITUTE(text, old, new)	=SUBSTITUTE("T1", "1", "2") -> "T2"
TEXT	=TEXT(value, format_string) Converts numbers/dates to text display.	=TEXT(TODAY(), "DD-MMM-YYYY") =TEXT(1250, "#,##0") -> "1,250"

PART 6: LOOKUP & REFERENCE FUNCTIONS

1. VLOOKUP (VERTICAL LOOKUP - TRADITIONAL STANDARD):

Syntax: =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
- lookup_value : Value to search for (MUST reside in Column 1 of table).
- table_array : The data range holding the lookup table (e.g. A2:M51).
- col_index_num : Column number to return data from (e.g. 2 for Airline).
- [range_lookup]: ALWAYS specify FALSE (or 0) for exact match!

Example: =VLOOKUP("FL-105", A2:M51, 2, FALSE)

* Limitation: VLOOKUP CANNOT look to the left of the lookup column!

2. INDEX + MATCH (THE TWO-WAY LOOKUP GOLD STANDARD):

- =MATCH(lookup_val, lookup_range, 0) : Returns the exact row position index.
- =INDEX(return_range, row_num) : Retrieves data from that row index.

Combined Formula:

=INDEX(B2:B51, MATCH("FL-105", A2:A51, 0))

* Superpower: Can look left or right, and never breaks when columns are added!

3. XLOOKUP (MODERN STANDARD IN EXCEL 365 & GOOGLE SHEETS):

Syntax: =XLOOKUP(lookup_val, lookup_range, return_range, [if_not_found])
- Defaults to EXACT MATCH automatically (no FALSE needed!).
- Built-in error handling via [if_not_found] argument.

Example: =XLOOKUP("FL-105", A2:A51, B2:B51, "Flight Not Found")

PART 7: DATE & TIME FUNCTIONS

FUNCTION	SYNTAX & DESCRIPTION	EXAMPLE
TODAY()	=TODAY()	Returns current system date.
NOW()	=NOW()	Returns current date and time.
DATE	=DATE(year, month, day)	=DATE(2026, 10, 24)
YEAR	=YEAR(serial_date)	=YEAR(TODAY()) -> 2026
MONTH	=MONTH(serial_date)	=MONTH(TODAY()) -> 10
DAY	=DAY(serial_date)	=DAY(TODAY()) -> 24
HOUR	=HOUR(time_val)	=HOUR(NOW()) -> 13
MINUTE	=MINUTE(time_val)	=MINUTE(NOW()) -> 45
DATEDIF	=DATEDIF(start_date, end_date, unit) Units: "Y" (years), "M" (months), "D" (days).	=DATEDIF(A2, B2, "D")
NETWORKDAYS	=NETWORKDAYS(start_date, end_date, [hols])	Calculates working business days.
EDATE	=EDATE(start_date, months)	Adds or subtracts n months.
EOMONTH	=EOMONTH(start_date, months)	Returns last day of the month.

PART 8: GOOGLE SHEETS DYNAMIC ARRAY SUPERPOWERS

1. UNIQUE FUNCTION (AUTOMATIC DEDUPLICATION):
Syntax: =UNIQUE(range)
Example: =UNIQUE(B2:B51)
Spills a clean, distinct list of unique airlines automatically without duplicates!

2. SORT FUNCTION (DYNAMIC TABLE SORTING):
Syntax: =SORT(range, sort_column_index, is_ascending)
Example: =SORT(A2:M51, 12, FALSE)
Dynamically sorts entire flight dataset by Delay_Mins (Col 12) descending.

3. FILTER FUNCTION (DYNAMIC LIVE SUBSETS):
Syntax: =FILTER(range, condition1, [condition2, ...])
Example: =FILTER(A2:M51, K2:K51="Delayed", C2:C51="T1")
Extracts all delayed T1 flights into an adjacent table with zero manual work.

4. ARRAYFORMULA (ELIMINATES MANUAL FORMULA DRAGGING):
Syntax: =ARRAYFORMULA(formula_applied_to_range)
Example: =ARRAYFORMULA(I2:I51 / H2:H51 * 100)
Computes Load Factor across all 50 rows in one single formula cell!

5. QUERY FUNCTION (SQL DATABASE QUERIES INSIDE SPREADSHEETS):
Syntax: =QUERY(data, "SELECT ... WHERE ... GROUP BY ...")
Example: =QUERY(A1:M51, "SELECT B, SUM(I) WHERE C='T1' GROUP BY B")
Executes full database aggregations natively in Google Sheets.

6. SPARKLINE & IMPORTRANGE:
- =SPARKLINE(range, {"charttype","bar"}) : Miniature in-cell chart.
- =IMPORTRANGE("sheet_url", "Raw_Data!A1:M50") : Live cross-sheet data sync.

PART 9: PIVOT TABLES & ADVANCED DATA ANALYSIS CHEATSHEET

PIVOT TABLE QUADRANTS EXPLAINED:

1. ROWS   : Categorical field whose unique values form the vertical row labels.
2. COLUMNS : Categorical field whose unique values form horizontal column headers.
3. VALUES : Numeric field summarized using SUM, AVERAGE, COUNT, MIN, MAX.
4. FILTERS : High-level slicer dropdown to isolate subsets (e.g. Flight_Type).

CRITICAL PIVOT COMPUTATIONS:

- Sum of Values   : Aggregates total volume (e.g. Sum of Pax_Booked).
- Count of Records   : Total movements/rows (e.g. Count of Flight_ID).
- Average of Values   : Mean turnaround delay (e.g. Average of Delay_Mins).
- % of Row Total   : Shows relative distribution per airline (Punctuality OTP).
- % of Grand Total   : Shows market share across the entire airport dataset.

PART 10: REAL-WORLD AVIATION FORMULA RECIPES

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1. PASSENGER LOAD FACTOR (% OCCUPANCY):

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=ROUND((I2 / H2) * 100, 1) & "%"
```

Divides Pax_Booked by Pax_Capacity, rounded to 1 decimal with percent sign.

2. ON-TIME PERFORMANCE (OTP 15-MINUTE INDUSTRY BENCHMARK):

```
=IF(L2 <= 15, "On-Time (OTP)", "Delayed")
```

Standard DGCA/FAA benchmark: departures within 15 mins of schedule are OTP.

3. BAGGAGE CAROUSEL LOAD RATIO:

```
=ROUND(J2 / I2, 2)
```

Checked baggage pieces divided by boarded passengers.

4. TURNAROUND DELAY SEVERITY TIERING:

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=IFS(L2=0, "Clean", L2<=15, "Minor Buffer", L2<=45, "Moderate", TRUE, "Critical")
```

LABORATORY & EXAMINATION GOLDEN RULES:

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1. Always begin every formula with '='.
2. Lock lookup tables and constants using absolute reference (A1) via F4.
3. In VLOOKUP, remember to set range_lookup to FALSE for exact matches.
4. In SUMIFS and AVERAGEIFS, the sum/average range is the FIRST argument.
5. Use IFERROR() around divisions to prevent unsightly #DIV/0! errors.
6. In Google Sheets, submit work as .xlsx by using File -> Download -> Excel.

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